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Roll No

ES-220(AU/ME/MI)-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Material Science

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the three most common space lattices observed in metals? Explain each in brief.
b) Define a crystalline substance. How does it differ from an amorphous material?
2. a) What are the engineering materials? Give the classification of engineering materials.
b) How does the heat treatment for hardening of Al and Cu alloy differ from that of steel?
3. a) Explain Gibb's phase rule in detail as applied to equilibrium diagram.
b) List the various micro-constituents of iron-carbon system compare their properties and composition.
4. a) What is the powder metallurgy? Explain various processes of powder metallurgy.
b) Explain Vander waal's or molecular bonding.

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5. Explain equilibrium diagram of an eutectic system. How solidification characteristics of eutectic system is different from isomorphism system?
6. What is the meaning of 'holding or soaking' time in heat treatment? How does it affect the Properties of steel?

OR

What type of heat treatment would you give to a high carbon steel to be used as a tool?

7. Explain with neat sketch the various point defects and line defects in metal crystal.

OR

Explain the difference between slipping and twinning. How does twinning occur in metals?

8. Write short notes on any two of the following:
 - a) Classification in bonds in solids
 - b) Metallurgical coke
 - c) Principle of recrystallization and grain growth
